Adam Bawatneh

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Website | GitHub | LinkedIn

Research Interests

Social reasoning and Theory of Mind, mental-state representations, and complex systems simulations.
Current work explores social reasoning in simulated societies and AI across language and multimodal settings.

Education

University of Central Florida Orlando, Florida
Ph.D. in Computer Science Aug 2025 – May 2029 (expected); GPA: 4.00/4.00
University of Central Florida Orlando, Florida
B.S. in Computer Science Aug 2022 – May 2025; GPA: 4.00/4.00

Selected Publication

Adam Bawatneh*, Sagar Sapkota, Amrit Singh Bedi, Santu Karmaker, and Mubarak Shah.
"OmniToM: Benchmarking Theory of Mind in LLMs via Explicit Belief Modeling."
NeurIPS 2026, Evaluations & Datasets Track. Project | Benchmark

Research Experience

Graduate Research Assistant, University of Central Florida May 2026 – Present
Conduct AI research and contribute to government proposal development, including IARPA-focused proposals.
Undergraduate Teaching Assistant, University of Central Florida May 2023 – May 2025
Led labs and mentored students in C, Python, and Java courses serving more than 1,500 students.

Selected Research Projects

OmniToM: Social Reasoning Evaluation for LLMs Oct 2025 – Present

- Led development of a benchmark spanning 895 stories and coordinated quality review with 11 annotators across more than 1,100 hours.
- Tested commercial and open models via Azure OpenAI, Anthropic API, and Google Vertex AI.

CADS: Context-Aware Detection of Specialized Language Summer 2026 – Present
Mentored Valerie Lopez during the summer REU; the project continues in the lab.

- Developed and evaluated an NLP pipeline that combines complementary text signals to identify specialized language.
- Evaluated transfer across domains and languages to assess where the approach generalizes.

BIAS-101: Demographic Bias in Video Action Recognition Jan 2025 – Apr 2025
Screened 85,000+ videos, curated 2,000+ clips, and analyzed demographic representation in action recognition data.
DETECT: Real-Time Gaze and Facial-Signal Analysis Aug 2024 – Apr 2025
Led a 6-member team to build and deploy a real-time gaze and facial-signal system. Project evaluation reported 80%+ gaze-tracking accuracy.

Academic Service

Conference reviewer: NeurIPS, AAAI, ACL, and ACM Web.

Technical Skills

Research: Social reasoning, Theory of Mind, research design, benchmarking, experimental analysis
AI: Natural language processing, machine learning, language models, model evaluation, responsible AI
Programming: Python, C/C++, Java, JavaScript, Go, SQL
Model Platforms: Azure OpenAI, Anthropic API, Google Vertex AI, Hugging Face
Cloud and HPC: Microsoft Azure, Google Cloud, NSF ACCESS
Tools: Git/GitHub, Linux, Docker, PostgreSQL, REST APIs, LaTeX